3 Basic principles of separation

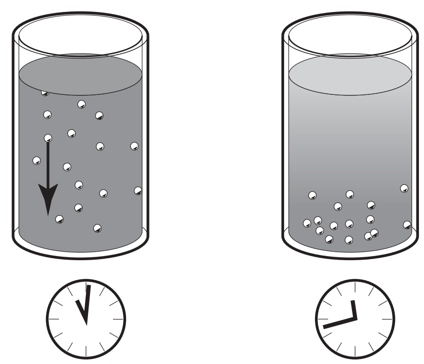
The purpose of separation can be to

- *free a liquid of solid particles,*
- *separate two mutually insoluble liquids with different densities while removing any solids present at the same time,*
- *separate and concentrate solid particles from a liquid.*

3.1 Separation by gravity

A liquid mixture in a stationary bowl will clear slowly as the heavy particles in the liquid mixture sink to the bottom under the influence of gravity.

A lighter liquid rises while a heavier liquid and solids sink.



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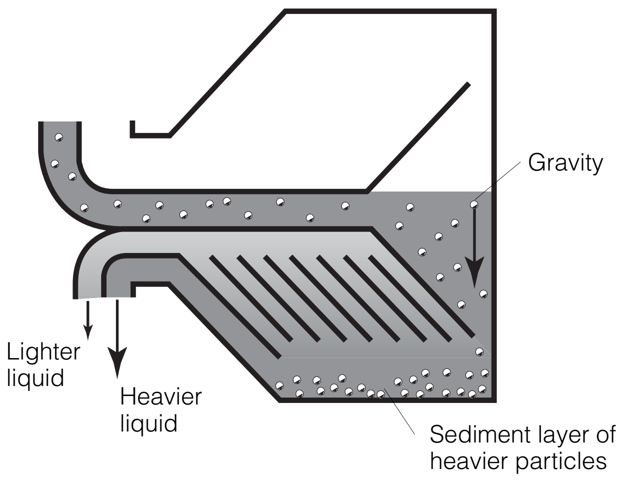
Continuous separation and sedimentation can be achieved in a settling tank having outlets arranged according to the difference in density of the liquids.

Heavier particles in the liquid mixture will settle and form a sediment layer on the tank bottom.



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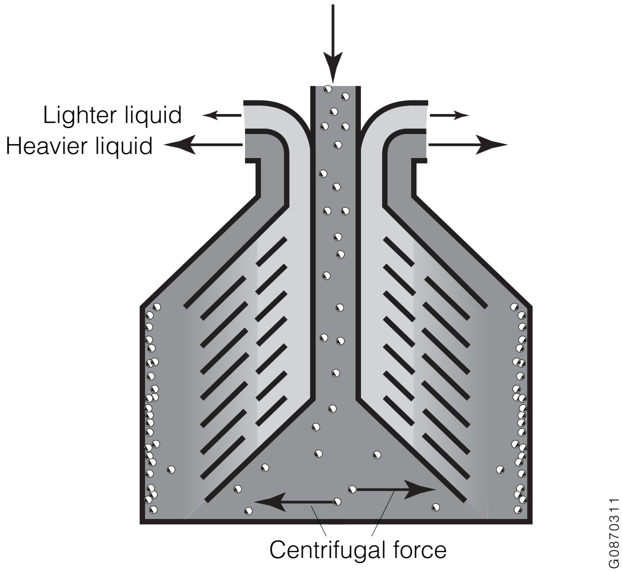
3.2 Centrifugal separation

In a rapidly rotating bowl, the force of gravity is replaced by centrifugal force, which is many times greater.

Separation and sedimentation is continuous and takes place very quickly.

The centrifugal force in the separator bowl can achieve in a few seconds that which takes many hours in a tank under influence of gravity.

The separation efficiency is influenced by changes in the oil viscosity, separating temperatures and in throughput.

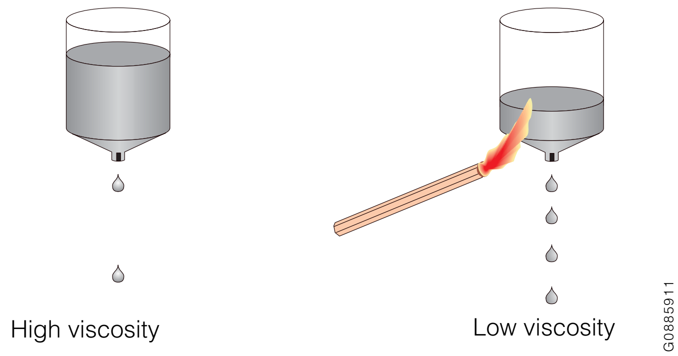


3.3 Temperatures

For some types of process liquids a high separating temperature will normally increase the separation capacity. The temperature influences oil viscosity and density and should be kept constant throughout the separation.

Viscosity

Viscosity is a fluids resistance against movement. Low viscosity facilitates separation. Viscosity can be reduced by heating.



Density difference

Density is mass per volume unit. The greater the density difference between the two liquids, the easier the separation. The density difference can be increased by heating.